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Alterations in subgingival microbiota during full-fixed appliance orthodontic treatment—A prospective study

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Abstract

Background: Full-fixed appliance orthodontic treatment (commonly called braces) increases plaque accumulation and the risk of gingivitis and periodontitis. However, little consensus exists on changes to subgingival microbiota and specific periodontopathogens during treatment with braces. Prior studies have been hampered by selection biases due to dependence on culture conditions, candidate-based PCR and shallow sequencing methods.

Objective: The objective was to provide the first longitudinal, culture-free and deep-sequence profiling of subgingival bacteria in subjects during early stages of full-fixed orthodontic treatment.

Methods: We performed 16S rRNA next-generation sequencing (NGS) on 168 subgingival samples collected at 4 distinct mandibular tooth sites per subject before (0 weeks) and during (6 and 12 weeks) orthodontic intervention in 9 experimental and 5 control subjects not undergoing treatment.

Results: Overall, we noted that orthodontic intervention led to increased microbial richness, accompanied by an increased incidence of localized gingivitis/mild periodontitis in subjects requiring orthodontic treatment compared to controls, as well as significant baseline variations in subgingival microbiomes in all subjects. Moreover, we confirmed individual- and site-dependent microbiome variability (in particular, the lingual site harboured higher microbiome diversity than buccal sites) that orthodontic bands may lead to more prolonged shifts in microbial changes compared to brackets, and evidence of adaptive enrichment of consensus bacteria with orthodontic intervention (12 novel, consensus bacterial species were identified).

Conclusion: Our study, along with evolving global profiling methods and data analyses, builds a strong foundation for further analyses of subgingival microbiomes during full-fixed orthodontic treatment.

KEYWORDS

16S rRNA, high-throughput nucleotide sequencing, oral microbiome, orthodontics

Isaac Chen and Jennifer Chung contributed equally to this paper.

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1 | INTRODUCTION

The oral microbiome is estimated to include at least 800 different species including a wide variety of gram-positive and gram-negative bacteria[1] and maybe in the thousands with the application of next-generation sequencing (NGS) techniques.[2] The oral microbiota colonizes sub- and supra-gingival surfaces of teeth, mucosa and saliva and experiences dynamic changes because of continuous exposure to the external environment.[3] Saliva contains ~1×10^9 bacteria/mL, whereas dental plaque has an equivalent bacterial concentration as the colon—1×10^{11} bacteria/mL.[4] The oral microbiome may be a key source in the aetiology of many oral and systemic diseases,[5,6] including dental caries and periodontitis,[7] diabetes,[8] infective endocarditis,[9] bacterial pneumonia,[10] adverse pregnancy outcomes,[11] inflammatory bowel disease,[12] pancreatic cancer[13] and colon cancer.[14] Despite the substantial impact on overall human health, the complexities of the oral bacterial communities and their compositional changes in response to environmental shifts are not well understood.[15]

The composition of subgingival microbiota may be influenced by several factors including oral hygiene, systemic disease, dental restorations and orthodontic appliances.[15] Full-fixed orthodontic treatment (brackets and bands commonly called braces) may change the subgingival environment by increasing plaque accumulation, levels of periodontopathogens, and deepening gingival sulci leading to gingival inflammation (bleeding, swelling, hyperplasia) and ultimately periodontitis.[16,17] Relative to misaligned, crowded teeth, aligned teeth may be more readily cleaned, which may benefit the periodontal condition.[18,19] In contrast, several studies have reported no significant differences and even decreases in the levels of periodontopathogens due to metal corrosion of orthodontic appliances imposing toxic effects on specific microorganisms.[20] In short, there is no or little consensus on adaptive changes of periodontopathogens in response to orthodontic treatment, if one does indeed exist. Prior studies on fixed orthodontic appliances and subgingival microbiomes utilized bacterial cultures, candidate-based PCR methods and Sanger sequencing, which suffer from restricted growth of bacterial cultures (less than half of subgingival bacterial species present can be cultivated using anaerobic microbiological methods),[21] assays for only select microbial species and shallow sequencing data respectively.[20]

The purpose of this study was to longitudinally and globally profile short-term, subgingival bacterial changes in subjects undergoing fixed orthodontic treatment using 16S rRNA PCR proceeded by NGS. We profiled 4 distinct sample sites in each subject at 3 time points (before orthodontic intervention; 6 and 12 weeks after intervention). To our knowledge, our study is the first to longitudinally analyse subgingival microbiota directly from sterile paper points without bacterial cultures at high resolution before and during full-fixed orthodontic treatment. We generated large data sets of subgingival microbial profiles that will clarify short-term changes in subjects treated with braces and build a foundation for future experiments.

2 | MATERIALS AND METHODS

2.1 | Subjects

Fourteen subjects (9 females and 5 males; ages 12y9m to 33y3m) were recruited and completed our study from the orthodontic clinic at UCSF. Nine full-fixed orthodontic patients (5 females, 4 males; ages 14 years 9 months ± 1 year 5 months) and five control/non-treated subjects (4 females, 1 male; ages 29 years 4 months ± 3 years 4 months) completed the 12-week study. The study design was approved by the UCSF Ethics Committee (IRB 15-17868). Subject inclusion/exclusion criteria were (a) no known systemic disease; (b) no use of antimicrobial, antifungal or anti-inflammatory drugs within 3 months before baseline examination; (c) no craniofacial anomalies; (d) not pregnant; (e) no active periodontal disease; (f) subject would be undergoing treatment with fixed orthodontic appliances in the mandibular arch; (g) and tooth crowding in the mandibular/maxillary arches was mild (<3 mm) so that extractions were not required (tooth spacing was also <3mm).

2.2 | Orthodontic bonding

All subjects received oral hygiene instructions from the initial consult in the form of verbal instructions and an instructional video (Aquarium 3, Dolphin Imaging & Management Solutions, Patterson Technology). Subjects were instructed to follow their normal routine and to not brush their teeth prior to appointments. Orthodontic bonding/banding is described in Appendix S1.

2.3 | Periodontal assessment

Immediately after sampling the sites, probing depth (PD), gingival recession (GR), clinical attachment level (CAL) and bleeding on probing (BOP) were measured for each site and recorded in the Axium periodontal charting system. A sterilized periodontal probe was used to measure PD and GR, CAL was calculated, and BOP was observed and classified as yes or no after the periodontal measurements (Table S1).

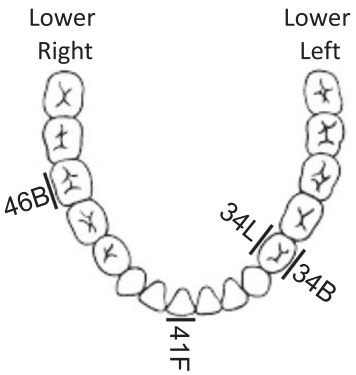
2.4 | Sample collection, 16S rRNA gene sequencing, data processing and data availability

Subgingival crevicular fluid samples were collected from 4 sites as illustrated in Figure 1 before subjects brushed their teeth. DNA from each sample was extracted using Qiagen DNeasy Powersoil kit. V1-3 region of 16S rRNA was sequencing on MiSeq platform using v3-600 cycle reagents (Illumina). DNA extraction and PCR amplification negative controls were included in the study to determine the potential contamination of NGS. Data processing was done based on standard protocol of USEARCH.[22] Taxonomic species classification

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(A)

Sample Site	Tooth	Fixed Appliance
46B	46 buccal	Band
41F	41 facial	Bracket (facial)
34B	34 buccal	Bracket (buccal)
34L	34 lingual	Bracket (buccal)



(B)

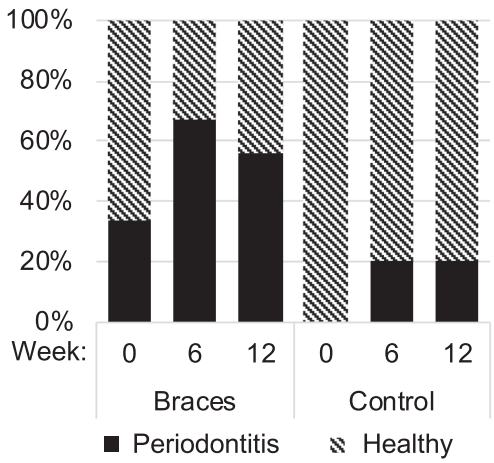


FIGURE 1 A, Table and diagram of sampled sites. B, Percentage of subjects with evidence of localized gingivitis/mild periodontitis (also refer to Table S1)

was identified via Human Oral Microbiome Database using BLAST.[23] Taxa that are likely potential contaminants were removed from the samples for downstream analysis. All sequencing data are available at the short reads archive (SRA) database with the access ID PRJNA552608. To summarize, we profiled 4 distinct sample sites in each subject at 3 time points (before orthodontic intervention, and 6 and 12 weeks after intervention).

2.5 | Statistical analysis

Subsampling of sequencing reads (read depth = 9367/sample) was performed at each sample to allow for microbiome comparison at the same read depth. Microbiome diversity was determined by Richness and Shannon Diversity as previously described.[24] Microbiome variation between subjects was illustrated by non-metric multidimensional scaling (NMDS). Permutational multivariate analysis of variance (PERMANOVA) was used to determine the statistical difference of the overall microbiome composition and abundance between groups. Differences in the relative abundances of specific taxa between the groups were determined with Deseq2.[25] The results from DESeq2 were further speculated by plotting the raw and relative abundance data. Results that are likely driven by outliers were removed from final reporting. Differences in Richness and Shannon diversity were analysed with Kruskal-Wallis testing for multiple group comparisons and followed by Wilcoxon rank-sum testing for the two-sample test. A P-value ≤.05 was considered statistically significant. All statistical analyses were performed in R (version 1.1.383).

3 | RESULTS

3.1 | Overview of demographics data and sequence data

Subgingival samples were obtained at 4 different sampling sites from 9 full-fixed orthodontic patients (braces) and 5 controls at 3 time points immediately before orthodontic intervention (0 weeks or baseline), and 6 and 12 weeks after delivery (Figure 1A). Indicators of periodontal health such as gingival PD, GR, CAL and BOP (Table S1). In the control group, none showed signs of compromised periodontal health at baseline, but one subject did at 6 and 12 weeks. In the braces group, one subject showed signs of compromised periodontal health at baseline, 7 at 6 weeks and 6 at 12 weeks (Figure 1B; Table S1), which corroborates prior studies regarding diminished periodontal health with orthodontic appliances.[18,26-28]

16S rRNA sequencing generated a total of 3 910 158 sequences after quality filtering with an average of 23 274 ± 10 689/per sample from 168 samples. One subject (Patient F- braces) yielded few reads and was not included in our analysis. Taxonomical classification revealed 417 unique operational taxonomic units (OTUs) from

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128 genera. *Streptococcus* was the most abundant genera across the 4 collection sites although buccal sites appeared to harbour more *Streptococcus* than lingual and facial sites. The 15 most abundant genera collectively made up on average 91.5% ± 7.8% of each sample taken from baseline (Figure 2A).

3.2 | Different sites harbour different microbiomes

To first examine microbiome complexity at different sampling sites, we compared alpha diversity at baseline. Alpha diversity was measured by Richness (calculated by the different number of OTUs present in a sample) and Shannon diversity (a measure of diversity that takes into consideration richness and evenness). Richness and Shannon diversity analysis showed that site 34L harboured significant complexity compared to buccal sites, 34B and 46B (Figure 2B). To determine variation between sampling sites, we performed non-metric multidimensional scaling (NMDS) analysis based on Bray-Curtis dissimilarity (Beta diversity) that accounts for composition and relative abundance of bacterial species (Figure 2C). NMDS confirmed that buccal sites (34B, 46B) were similar to each other but distinct from lingual (34L) or facial sites (41F) (P =.003, PERMANOVA). The relative abundance of *Streptococcus* at buccal sites was almost twice that of the lingual site (Figure 2A). Samples from 41F showed variation between buccal (34B, 46B) and lingual (34L) sites.

3.3 | Increased alpha diversity of subgingival microbiome with orthodontic intervention

Next, we determine changes in alpha diversity before and during treatment. Richness was not significantly different in controls at 0 vs 6 weeks and was slightly increased at week 12 (Figure 3A, middle panel), suggesting that the number of bacteria is generally stable over time under non-orthodontic conditions. In contrast, microbial diversity in subjects with braces at 6 and 12 weeks was significantly higher compared to baseline at all sites (Figure 3A, middle panel; red asterisks). Specifically, orthodontic intervention resulted in the emergence of 42 new bacterial species at week 6 from all sites collectively (data not shown)—on average, accounting for 10.81% of total abundance. Each site demonstrated significant Richness differences between 0 and 6 weeks, and differences were maintained at 12 weeks in all combined sites of the braces group, suggesting a sustainable dysbiosis of subgingival microbiomes once disrupted by orthodontic intervention. Shannon diversity showed trends similar to Richness—increases at 6 and 12 weeks relative to baseline in subjects with braces group for overall sites, but no significant differences except for 46B (Figure 3A, right panel; red asterisks), highlighting site-dependent adaptive changes. Comparisons between control and braces groups showed significantly higher or increasing trends of Richness and Shannon diversity with braces at 6 and 12 weeks (Figure 3A; purple asterisks) suggesting global overgrowth of bacteria in the short term across different sites.

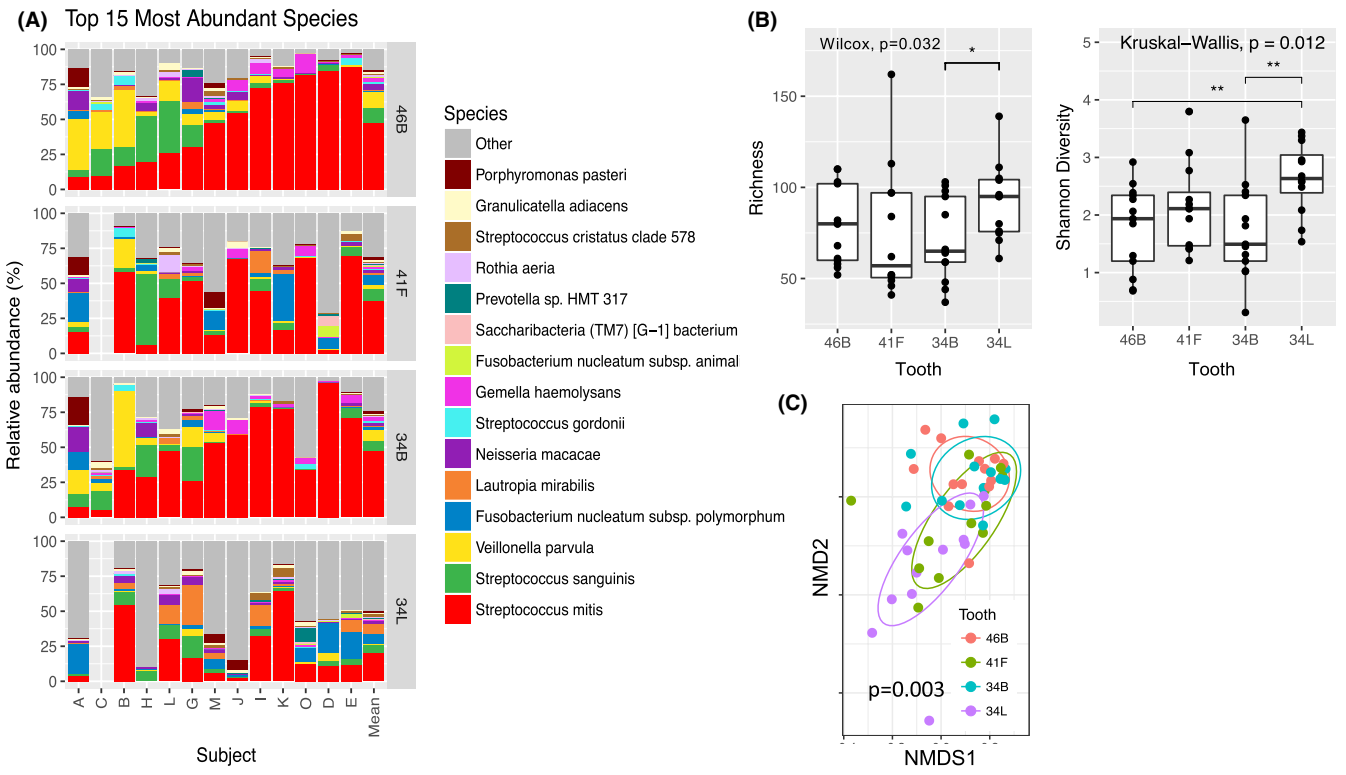


FIGURE 2 Microbiome differences across different collection sites at baseline. A, Top 15 most abundant species in each individual participant at four collection sites at baseline week 0. B, Microbiome diversity comparison at different collection sites. C, Overall microbiome community comparison by NMDS. The microbiome is significantly different from buccal sites (P = .003 by Adonis)

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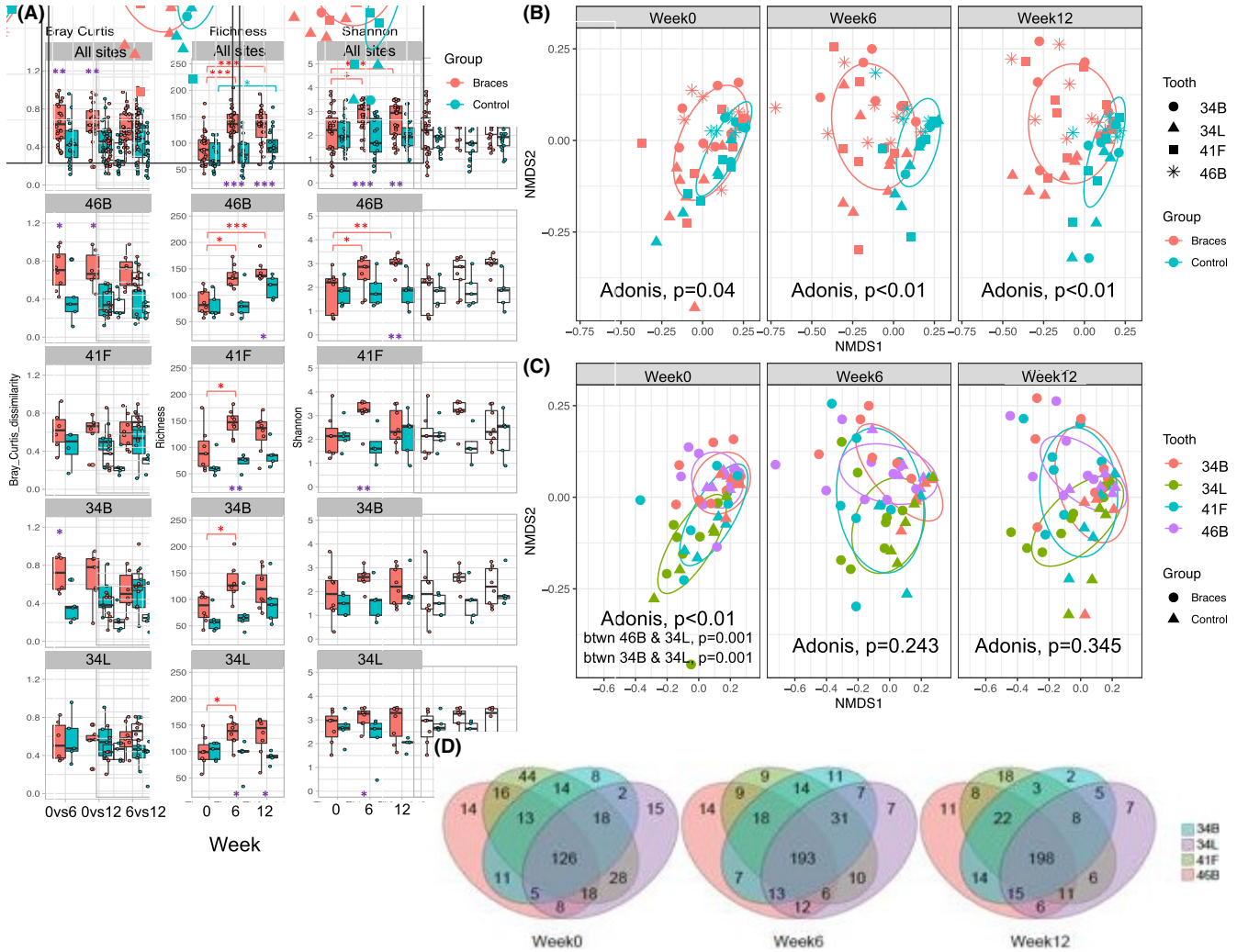


FIGURE 3 Alteration of the microbiome from baseline to 6 and 12 weeks after braces intervention. A, Change of alpha and beta diversity. Beta diversity is calculated by Bray-Curtis dissimilarity for different time points for braces or control group. Statistical analysis is performed to compare the changed beta diversity between brace and control. Significant values are indicated by purple asterisks (*P <.05, **P <.01, ***P <.001). Richness and Shannon diversity comparison is done at each time point between braces and controls, and the statistical significance is indicated by purple asterisks. Richness and Shannon diversity comparison is also done at different time points for either braces or control (red or green asterisks). Comparisons without statistical differences are not indicated. B, Significant differences of the microbial communities between braces and control at baseline, 6 weeks and 12 weeks. C, Decreased microbiome difference across sample collection sites after braces intervention. D, Shared taxa are increased after braces intervention

3.4 | Changes of beta diversity of subgingival microbiome with orthodontic intervention

Beta diversity, generally quantified by Bray-Curtis dissimilarity, measures differences in microbiomes. The overall microbiome communities between braces and controls at baseline were significantly different (Figure 3A,B), which raises the possibility that malocclusion may influence microbial composition. Six and 12 weeks of braces also showed significantly different microbiomes compared to controls (Figure 3A). The increased beta diversity after orthodontic intervention was particularly evident in buccal sites (46B, 34B), even though 41F also showed a trend of increased beta diversity in the braces group (Figure 3A). The overall microbiome communities were significantly different between the two groups at 6 and 12 weeks (Figure 3B) highlighting overt microbial community alterations with orthodontic intervention.

Because we demonstrated that niche-specific microbiomes exist at baseline, we next asked whether orthodontic intervention altered this ecological microbiome configuration. Interestingly, PERMANOVA analysis revealed that microbiomes at the 4 sampling sites were not statistically different at 6 (P =.243) and 12 (P =.345) weeks (Figure 3C) and coincided with an increase in the number of shared OTUs at 6 and 12 weeks (Figure 3D). Thus, orthodontic intervention appears to initially disrupt established microbiome niches then trends microbial communities to be more homogenous and less distinguishable.

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3.5 | Changes of specific taxa after orthodontic intervention

To identify microbial changes, we compared microbiomes between the 3 time points. Although overall microbiome structure in the control group was maintained from 0 to 12 weeks, we identified specific OTUs that were significantly altered between 0 to 6 and 0 to 12 weeks (17 and 53 OTUs respectively). This may reflect the natural fluctuation of the subgingival microbiome within our study time frame. To identify taxa changed by orthodontic intervention, these naturally fluctuating taxa identified in controls were removed from the taxa list that was significantly different from orthodontic intervention. We identified a large number of taxa that were significantly altered at 6 and 12 weeks with braces compared to baseline (Figure 4B, grey and pink bars respectively). We found similar microbiomes at 6 and 12 weeks after orthodontic intervention, indicating that the influence of braces on subgingival microbiomes persists to 12 weeks. Again, buccal sites displayed more profound changes at 12 weeks, as indicated by the large number of OTUs enriched at 46B and 34B sites. This observation suggests that microbiomes will continue to change after 12 weeks, even though these changes were not statistically significant at 6 weeks (Figure 4B).

The cross-sectional comparisons between braces and control groups at time points, as well as longitudinal comparisons within the braces group, resulted in 24 comparisons in total. We identified 12 OTUs that were consistently enriched in subjects with braces compared to controls at multiple sites and time points, as well as at 6 and 12 weeks compared to baseline in subjects with braces (Figure 4C). Consistent enrichment was defined as enrichment in at least 8 comparisons among the total 24 comparisons. Out of the 12 bacterial species that were significantly enriched, *Selenomonas sputigena* showed the most consistent response to orthodontic intervention across different sites. Our results demonstrate that specific taxa in different niches adapt to orthodontic treatment in a similar fashion, whereas niche-specific adaptations were also evident as not all taxa showed changes across all sites.

4 | DISCUSSION

To date, little consensus exists on the composition of healthy oral microbiomes and adaptive microbial changes that are associated with braces. Moreover, it is unclear how different ecological niches such as different teeth in the same individual or even different sampling sites within the same tooth respond to orthodontic treatment. In our study, 4 distinct sites per subject were collected before and during orthodontic intervention to understand niche-specific microbiota diversity including differences between posterior (46 and 34) vs. anterior teeth (41), buccal (46B, 34B) vs. lingual (34L) sites and banded (46) vs. bracketed (34, 41) teeth. Numerous tooth sites of each subject were assayed to catalog the differences in initial microbial composition and its changes with and without orthodontic intervention. To our knowledge, our study is the first to longitudinally analyse subgingival microbiota using 16S rRNA-based NGS, directly from sterile paper points, without bacterial cultures, at high resolution before and during full-fixed orthodontic treatment. A prior study utilizing 16S rRNA NGS in participants wearing braces was focussed on supra-gingival microbiota on a single tooth and the effects of fluoridated mouth rinse on the microbiome.[29]

Despite some evidence for a conserved healthy oral community, prior studies have shown large inter-individual differences in healthy individuals in supra- and subgingival bacteria[30] and saliva.[31] Bik et al (2010) profiled 10 healthy mouths by pooling 26 specimens from 7 tooth sites, but the study did not note any dental crowding or need for braces. In our study, control subjects that did not require orthodontic treatment presented with no or little dental crowding or spacing (0-1mm) in either arch, whereas braces subjects showed between 3 and 7mm of crowding (data not shown). Braces subjects prior to orthodontic intervention harboured characteristic microbiota distinct from controls (Figure 3B), which supports the observation that there may be a correlation between malocclusion (such as dental crowding) and periodontal disease.[18] Moreover, within the braces group, greater inter-individual and inter-site variations of microbiome compositions were observed (Figure 3B-C). Site-specific variation was evident in lingual compared to buccal sites, which harboured less *Streptococcus* and a greater number of diverse bacteria (Figures 2B-C, 3B).

Upon orthodontic intervention, our data showed adaptive shifts in microbiota at 6 and 12 weeks compared to controls (Figure 3A,B). However, within the braces group, microbial differences notably decreased from 6 to 12 weeks indicated by NMDS plots (Figure 3A)—a trend supported by increases in shared species in all sites at 6 and 12 weeks (Figure 3D). These results suggest that orthodontic intervention shifts the microbiota towards specific species and offer a possible explanation as to why certain periodontopathogens decreased during treatment.[32]

The only banded tooth site (46B) exclusively showed significant differences between week 0 and week 12 (Figure 4B). Thus, orthodontic bands may lead to prolonged retention of microbial changes, whereas brackets may lead to transitory changes, supporting the observation that the use of orthodontic bands is associated with clinical attachment loss.[33] However, additional studies with increased subject number and treatment times are required.

The relatively small number of individuals sampled was a limitation of our study. However, the large inter-individual variability in microbiomes[30] strongly supports the use of longitudinal comparisons (rather than increasing the number of individuals) to identify consensus enriched species due to orthodontic treatment. Although we profiled 5 control and 9 braces subjects (one subject was omitted due to low reads), our experimental design resulted in 60 (5 subjects x 4 sites x 3 time points) and 96 (8 x 4 x 3) distinct sequencing profiles from control and braces subjects respectively. If we consider each specific site, our experiment essentially resulted in 20 control and 32 braces longitudinal sets of data. We identified 12 enriched consensus species with orthodontic intervention (Figure 4C; 2 sub-species of *Selenomonas*

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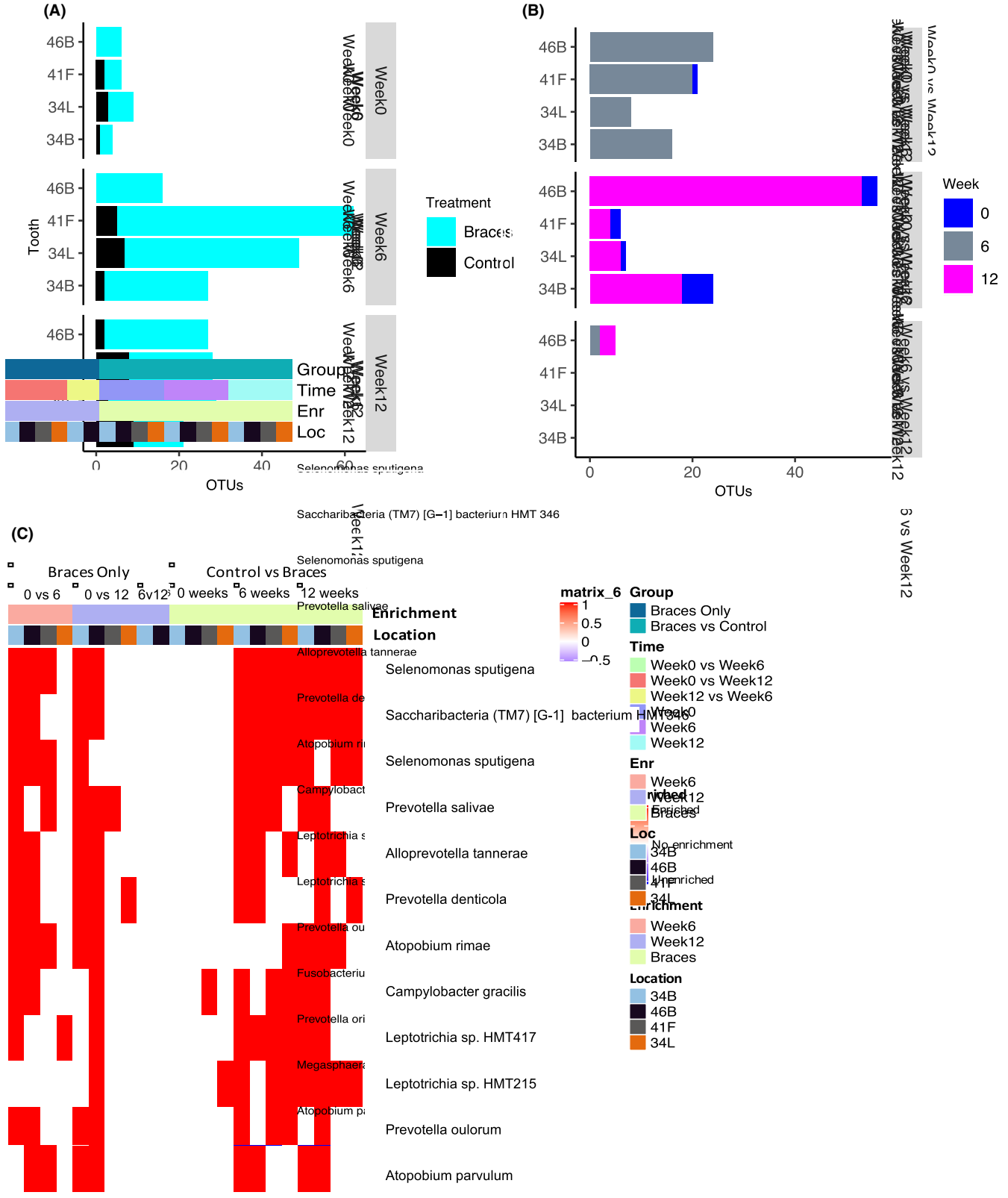


FIGURE 4 Microbiota alteration after brace intervention. A, The number of taxa that are significantly different between brace and control groups at different time points. B, The number of taxa that are significantly altered in brace group after brace intervention C, Specific taxa that are significantly enriched across different comparisons (brace and control comparison at different time points and different time points comparison after brace intervention at brace group. Taxa enriched at least from 8 comparisons are plotted by heat map

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sputigena were noted). None of the species identified have previously been reported to be enriched with orthodontic treatment. *Selenomonas sputigena*, *Campylobacter gracilis* and *Leptotrichia* have been associated with periodontitis previously[20,34,35] but the remaining species, namely *Saccharibacteria* (TM7), *Prevotella denticola*, *Prevotella salivae*, *Atopobium rimae*, *Alloprevotella tannerae*, *Prevotella oulorum* and *Atopobium parvulum*, were not associated with periodontitis. *Saccharibacteria* (TM7) are ubiquitous members of the human oral microbiome that have recently been identified through deep-sequencing and appear to play a role in oral mucosal infectious diseases[36]; *Prevotella intermedia* (*Pi*) is a commonly reported periodontopathogen (see below), but *Prevotella denticola* and *Prevotella salivae* were neither associated with orthodontic intervention nor periodontitis; *Atopobium rimae* may be associated with periodontal abscess formation[37]; and *Prevotella oulorum* was associated with experimental gingivitis.[38] Further functional characterization of these additional bacteria, in particular *Selenomonas sputigena*, will improve our understanding of the impact of orthodontic intervention on oral health.

Our study highlights the complexity of cataloging subgingival microbiota due to large microbiome diversity, the limitations of cell culture methods, candidate-based PCR and shallow sequencing approaches of prior studies. With the adoption of new technology, we will need to re-investigate subgingival bacterial compositions. For example, 4 periodontopathogens are commonly assayed in studies of microbial changes during orthodontic treatment—*Aggregatibacter actinomycetemcomitans* (*Aa*), *Porphyromonas gingivalis* (*Pg*), *Pi* and *Tannerella forsythia* (*Tf*).[20] In a prior systematic review involving these 4 bacterial species, only *Tf* increased significantly between 0 and 12 weeks of treatment and *Pi* significantly increased in the incisors but not the first molars.[20] In our study, *Tf* increased significantly after 6 and 12 weeks of orthodontic intervention in both molar (46B) and incisor (41F) (data not shown). However, *Aa* and *Pi* showed no changes and *Pg* was too low in abundance to detect (data not shown). Our results demonstrate that orthodontic-induced periodontitis may involve different bacterial species and highlight the need for further research in understanding compositional changes in oral microbiota during full-fixed orthodontic treatment.

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CONFLICT OF INTEREST

The authors declare that there are no conflicts of interest.

AUTHOR CONTRIBUTIONS

IC contributed to conception and design, data acquisition and analysis and drafted the manuscript; JC and RV contributed to data analysis and interpretation, performed statistical analyses, drafted and critically revised the manuscript; GW contributed to analysis and critically revised the manuscript; YZ and AHJ contributed to conception, design; data acquisition, analysis and interpretation; drafted and critically revised the manuscript. All authors gave their final approval and agreed to be accountable for all aspects of the work.

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